UNIT42 · FIRMWAREX

Firmware Security Assessment

MIPS controlled E2E firmware

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Executive summary

This assessment recorded **16 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	9	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	MIPS	Not detected	Not detected	none	Present	Not detected
/bin/taint-demo	MIPS	Not detected	Not detected	none	Not detected	Not detected

Detailed findings

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1. Command injection candidate High

Property	Result
CWE	CWE-78
Affected file	/bin/taint-demo
Architecture	MIPS
Confidence	probable
Review status	new
Address	0x500004
Function / sink	system
Detected	2026-10-20T05:30:45.045213Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x400680 → 0x500000 → 0x4006cc → 0x4006e8 → 0x500004
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args … [excerpt; full expression in JSON export]
stack hash	e5d33603a1ee68b5b85a27213e73c3a8a502862e8f2cbfe90b4fc88e5bb6215b

Evidence reference: evidence:d79ec72c4f50280a86a1e71c0f75e1892aff74a88a9ecdee8a0924a5eda0238d. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review whether external input can reach command execution. Prefer structured APIs and strict allowlists; validate the candidate before remediation.

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2. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172611Z

Evidence

Observation	Value
rule	credential
fingerprint	80105497342cfdfe6669b979adc225185654b625cc58a9e39cd7de3389433eaa
byte offset	9
byte length	19

Evidence reference: evidence:d8de469bc44b2b92db0e3a8850295d1f7828c4588b4d1588c011a328d4a29740. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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3. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172654Z

Evidence

Observation	Value
rule	credential

Observation	Value
fingerprint	e63b9dadcd4a050b1fc8f00541e5cbb2de4e1aea7181a688a75860d58574565f
byte offset	37
byte length	24

Evidence reference: evidence:963203996050768747b6774baa9017e5b7449959389ede013d13a0f7847cc2d7. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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4. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172682Z

Evidence

Observation	Value
rule	private_key
fingerprint	79bb74ebba1244fafad9364f262521355e9a7d3d300166c1c276659e2f008578

Observation	Value
byte offset	62
byte length	78

Evidence reference: evidence:f9a688ad6976464388fd5bf338882bf190938e73de51cf502381fade16189d79. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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5. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172775Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	13ae62efebf63ab85a91b9808b46ba90c9732e4960878b9216b64be4af0a0c08
byte offset	237
byte length	3

Evidence reference: evidence:58efb09f64cd09e07ebe8297d65bca23080d6d71c7fce8a4340664c44b3a6651. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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6. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172804Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	2898c701d7ba938e1adb0702949407397bcc62c63f0015c4380da1f70632f6c8
byte offset	255
byte length	3

Evidence reference: evidence:42d66f5e9b5778803e508732439f5dfe797a96591a2d5de8501814b1a93220b5. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-1b3fcbad97f68e26
Architecture	MIPS
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.173071Z

Evidence

Observation	Value
rule	credential
fingerprint	0539d9ce49715191af7bc6ee4f5ac6b05db53006544ec85f9686edabbb12f532
byte offset	38
byte length	25

Evidence reference: evidence:d8a8f6e36fec9ab1d75007b5c80311ec55c61420e49c0329acdc495173b1f59d. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

8. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-20T05:30:41.583515Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

9. Embedded secret candidate Low

Property	Result
CWE	CWE-798

Property	Result
Affected file	/bin/service-probe
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.171699Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	cb57b3a02ede6faccc4fdc6895b2b22d6600fb42f787c0810f180aa30a856ceb
byte offset	469114
byte length	50

Evidence reference: evidence:90b1c59cafe1dbe2f2ed7230765ab88bd0e83f9117475c31b9758017c72f7dd7. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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10. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPS

Property	Result
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.171780Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	044064cd7e5f801cffb69278e51520ca1281745e5557cb3543e92f3527a657a6
byte offset	548957
byte length	26

Evidence reference: evidence:e872451a50cf65f50fe6606394af0c10b52de8e9aaafdf78f0eda87d4e6d43f0. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPS
Confidence	heuristic
Review status	new

Property	Result
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.171820Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	280b0f1b68782684651586343 7e83f0b9617ae78f2851fc636cdff04b7f13ac8
byte offset	548989
byte length	26

Evidence reference: evidence:36032cde0142186d33bcaf330db4b90577ac8f74c0525785cde044aac45523a2. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable

Property	Result
Detected	2026-10-20T05:30:45.171848Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	a1c7fb691be20c86c23ca98ae99fd1e3dd6bdee069d1cef781d83aa7f61e2fde
byte offset	549856
byte length	36

Evidence reference: evidence:58894f852122d1db7b24bc94eb011fb4daba9931f143d06ddb7d7d780eae6534. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.171871Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	53b809746b8919dab0cdbf9f23726f7ae341f1c6f8c81db174572d0e1d85daf1
byte offset	549896
byte length	36

Evidence reference: evidence:a616b839ce312beab80e13e9664cc78c4917715aa47f2a6fc66e2fd353f8da6d. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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14. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172707Z

Evidence

Observation	Value
rule	certificate

Observation	Value
fingerprint	167f23504c372b88081993dcf7e7723607683d6dd1c3a6f080cbf6764625c597
byte offset	141
byte length	75

Evidence reference: evidence:964afb4079454e82758346ab08ca41983b0d9f027fb2962d0956cd5f716f15af. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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15. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172727Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	43451bc12369c7cf6e7e23f9a2921bdff1e4482f690fddfcc53672913d9cea4c

Observation	Value
byte offset	29
byte length	32

Evidence reference: evidence:a1771d5535b15a48c3a2538945e827ffd0334d53e4400d1455d2427cb1fb0b81. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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16. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	MIPS
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:45.172751Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	74fd6091433bf2c8c62056069b10b260b356734f1559a5ed63e2ffbcadd085cb
byte offset	259
byte length	29

Evidence reference: evidence:4d1118b8c1131717d4a3b3320502b85a9ea5ded5ed7fcdb20daea5a6ec9c7120. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	b29bf453-7cd5-4e24-9951-0700068a8872
SHA-256	b8b8d218b355f87c721ce5129ccc5623905c249e1e92565b9f4b69a45c68d255
Job ID	730c3904-b617-4617-a3dc-d4abd420e36e
Image bytes	315392